

CS 301: Machine Learning & RAG Architecture

Chapter 1: Introduction to Machine Learning and Neural Networks

Machine Learning (ML) is a branch of artificial intelligence that enables systems to automatically learn patterns from data without being explicitly programmed.

Supervised learning trains models on labeled datasets to predict continuous targets (regression) or discrete categories (classification). Unsupervised learning discovers hidden structures in unlabeled data through techniques like K-Means clustering and Principal Component Analysis (PCA).

Neural Networks consist of interconnected nodes (neurons) structured in input, hidden, and output layers. Weights are optimized via gradient descent and backpropagation.

Chapter 2: Natural Language Processing and Embeddings

Natural Language Processing (NLP) focuses on enabling computers to understand, analyze, and generate human text.

Embedding models (such as Sentence-Transformers) convert raw text into dense, continuous numerical vector representations in high-dimensional space. Semantically similar sentences map to nearby vector coordinates.

Cosine similarity measures the angle between two embedding vectors. A cosine distance near 0 indicates high semantic similarity, enabling efficient nearest-neighbor search in vector databases.

Chapter 3: Retrieval-Augmented Generation (RAG) Architecture

Retrieval-Augmented Generation (RAG) combines dense vector retrieval with Large Language Models (LLMs) to produce accurate, factual responses grounded in external domain documents.

The standard RAG pipeline operates as follows: 1) Document text extraction and chunking, 2) Dense vector embedding generation, 3) Persistence in a vector database like ChromaDB, 4) Top-K vector similarity retrieval for user questions, and 5) LLM response generation constrained by retrieved context.

RAG mitigates LLM hallucinations, eliminates expensive fine-tuning for dynamic knowledge, and provides transparent source attribution with document and page citations.